

Project Interim Report
On
Contact Directory Management
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming with Java

SESSION: 2025-26



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Abstract

The Contact Directory Management System is a Java-based project developed using Object-Oriented Programming (OOP) concepts. This system is designed to store, manage, and organize contact information efficiently. Traditional methods of maintaining contact details manually can lead to data loss, duplication, and difficulty in searching.

This project provides a digital solution where users can add, search, update, and delete contact details easily. Each contact contains information such as name, phone number, and email address. The system ensures quick access and better organization of contacts.

The main objective of this project is to demonstrate OOP concepts such as encapsulation, abstraction, inheritance, and polymorphism. It provides a simple and effective way to manage personal or professional contacts.

Introduction

The Contact Directory Management System is developed to simplify the management of contact information using Java programming and OOP principles. Managing contacts manually in notebooks or files can be inefficient and time-consuming.

This system allows users to store contact details digitally and perform operations such as adding new contacts, searching existing ones, updating information, and deleting unwanted contacts. It improves data organization and retrieval speed.

The project demonstrates how object-oriented programming can be used to solve real-life problems effectively.

Objectives

- Store contact details efficiently
 - Add new contacts
 - Search contacts by name or number
 - Update existing contact details
 - Delete contacts
 - Reduce manual work
 - Apply OOP concepts in Java
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Scope

This project is suitable for personal and small-scale use. It can be extended by adding database integration, cloud storage, and mobile application support.

System Requirements

Hardware:

- Computer with minimum 4GB RAM

Software:

- Java JDK
 - IDE (VS Code / Eclipse / IntelliJ)
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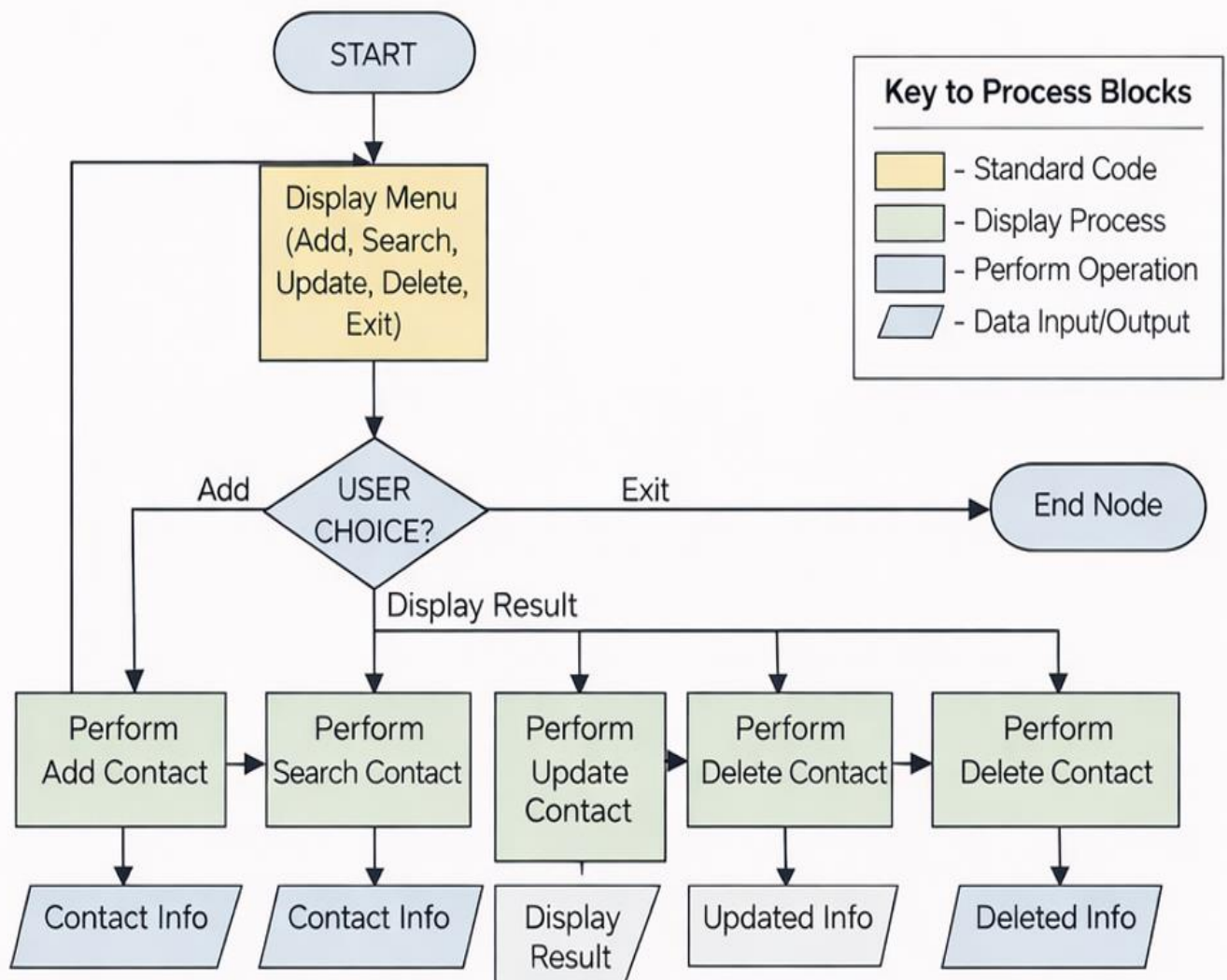
Methodology

1. Start the program

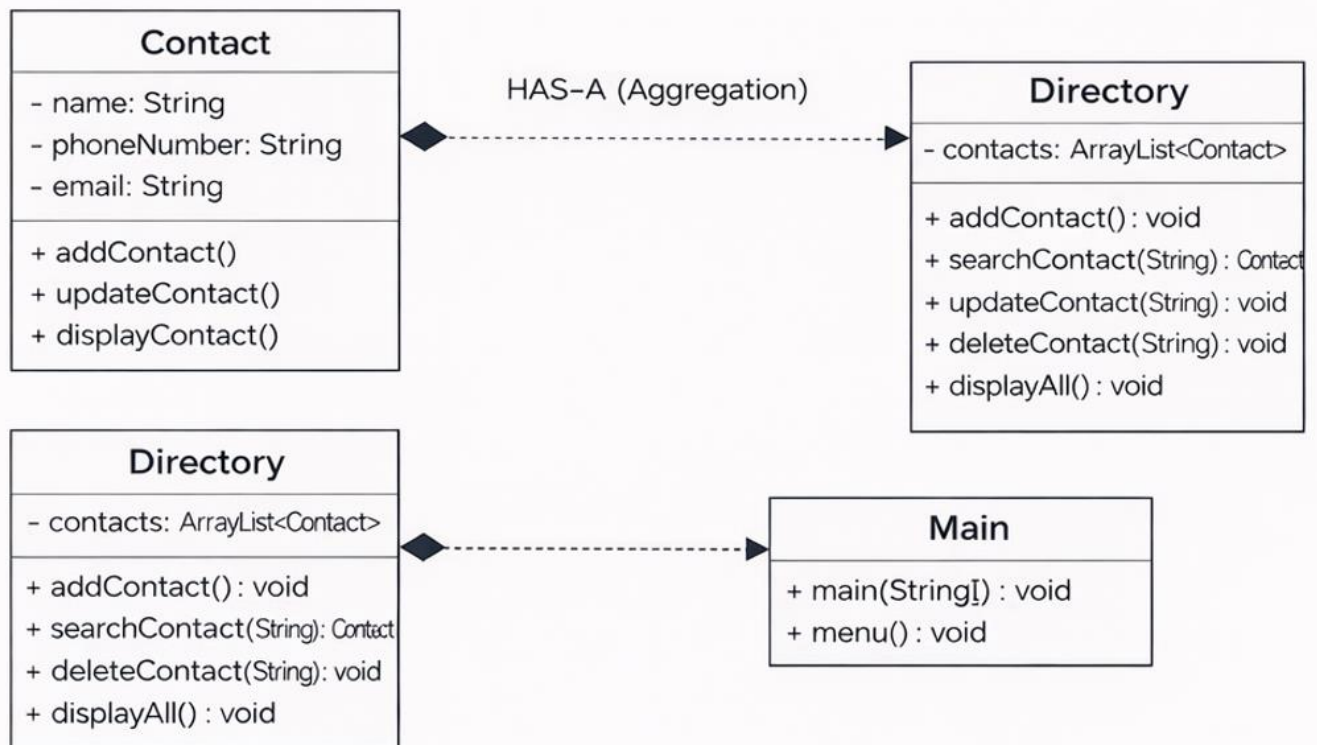
2. Display menu options
3. User selects operation:
 - Add Contact
 - Search Contact
 - Update Contact
 - Delete Contact
 - Display All Contacts
4. Perform selected operation
5. Store data using ArrayList or array
6. Display result
7. Repeat until exit

Flowchart

PROGRAM FLOW: Contact Directory Management System



Class Diagram



Implementation

- **Contact Class:** Stores contact details such as name, phone number, and email.
- **Directory Class:** Manages operations like adding, searching, updating, and deleting contacts.
- **Main Class:** Handles user interaction and menu display.

The system uses `ArrayList` to store contact objects and perform operations efficiently.

Sample Input/Output

1. Add Contact
2. Search Contact
3. Update Contact
4. Delete Contact
5. Display All
6. Exit

Output:

Contact Added Successfully!
Contact Found!
Contact Updated Successfully!
Contact Deleted Successfully!

Future Enhancements

- Database integration (MySQL)
- Contact import/export feature
- GUI using Java Swing
- Cloud synchronization

- Mobile application

Gantt Chart

Task	Duration
Planning	2 Days
Design	2 Days
Coding	5 Days
Testing	2 Days
Documentation	2 Days

Conclusion

The Contact Directory Management System successfully demonstrates the use of Java OOP concepts. It provides an efficient way to store and manage contact information digitally, reducing manual effort and improving accessibility.

References

Book:

[1] H. Schildt, *Java: The Complete Reference*, 11th ed. New York: McGraw-Hill, 2018.

Website:

[2] Oracle, "Java Documentation," Oracle. [Online]. Available: <https://docs.oracle.com>